



ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG QUỐC TẾ
VNU-INTERNATIONAL SCHOOL

EXAMINATION

Code: 01

Program: ICE Course Code: INS3118 Course Title: Engineering and Computer Graphics Level: Bachelor Time allowed: Date: Time: 10 minutes for video presentation and 5 minutes for Q&A	<i>Lecturer's Signature</i> Giáp Văn Tiến <i>Date:</i>
	<i>Department's Signature</i> <i>Date:</i>

Instructions to students:

1. Closed/Opened book examination: Opened books
2. Answers are to be written in the answer provided.
3. Students are NOT permitted to retain this examination paper

This exam paper contains 5 sheets, including the cover.

PROJECT REQUIREMENTS

1. Instructions:

- Each group should consist of maximum 3 students.
- Each group is expected to select UNIQUE PROJECT in Table 1 to DEVELOP following the requirements.

Table 1: Project titles:

No.	Theme	Room Layout Variation	Lighting Twist	Camera/Texture Twist
1	The Art Gallery	Long rectangular hallway.	Multiple dim point lights.	Must display 3 different images on 3 different walls.
2	The Night Watch	Square room with central pillar.	Light bulb "flickers" (random intensity).	CCTV view has a "Green Night Vision" shader effect.
3	The Mirror Maze	L-shaped room.	Light is a Spotlight attached to the head.	One wall must be "shiny" (high specular) to act as a mirror.
4	The Disco	Hexagonal room.	Light changes color over time (RGB cycle).	First-person view has a "fisheye" lens distortion or at least one furniture (table/chair)
5	The Laboratory	Two small rooms connected by a door.	One light in each room.	The image on the wall is "Semi-transparent" (Alpha Blending).
6	The Deep Sea	Circular room (Approximated by cylinders).	Light is blue/murky with high attenuation.	The CCTV camera "shakes" slightly as if underwater.
7	The Warehouse	Room filled with 3 static crates (cubes).	Light is placed as table lamp	Must use "Mipmapping" for the wall texture to prevent moiré.
8	The Space Station	Room with a "window" (transparent texture).	Light source is "Sunlight" (Directional Light).	The "Head" ball can fly up and down (Z-axis movement).
9	The Haunted House	Narrow corridor with a sharp turn.	Light follows a predefined path (moving light).	Wall texture is a "Portrait" that changes image when clicked.

10	The Neon Grid	Standard Room.	No bulb; the "Ball" itself is the light source.	The floor has a "Grid" texture instead of a solid color.
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- Project requirements: create an interactive graphics, that simulates a room with light source inside, the wall has picture/mirror on it. One ball moved as user pressing keyboard and first-person view changes direction by mouse. The view can be from a CCTV at the corner of the room, which can change the direction.
- All students are required to submit their full project materials, including compiled (build) folder to be run in Windows OS, source code, 3D model files, presentation files, via Google Drive **at least three days before the exam date.**
- Students are required to register their chosen problem for the final exam in the 7th week of the semester.

2. Report

The project report includes the following sections:

PROJECT STATEMENT

MEMBER CONTRIBUTION TABLE

CONTENT

LIST OF FIGURES

LIST OF TABLES

NOMENCLATURES

GLOSSARY

Chapter 1. INTRODUCTION

- 1.1. Introduction to project
- 1.2. Motivation
- 1.3. Aims and Objectives
- 1.4. Methodology
- 1.5. Report outline

Chapter 2 Geometry & Environment

- 2.1. 3D modeling
- 2.2. Texture mapping (image or mirror handling)
- 2.3. Asset loading

Chapter 3. Navigation & Viewports

- 3.1. Coordinate Spaces
- 3.2. The "head" ball controller
- 3.3. Dual camera toggle (between first person and third person view)

Chapter 5. Illumination & shading & shadow

- 4.1. Point Light
- 4.2. Shader
- 4.3. Attenuation

REFERENCES

APPENDIX

+ **Source code**

+ **AI prompt**

3. Presentation

- *Each group has 10 minutes to present and 5 minutes for Questions and Answers.*
- *Summarizing your work using MS **PowerPoint**, which should include the following parts:*

1) Introduction:

- Begin your presentation with an introduction.
- Provide context for your project.
- Mention the main objective of your presentation.
- Highlight any challenges or issues associated with the topic of your group.

2) Methods:

- Explain the methodology or approach you have taken to solve the problem.
- Describe the techniques, algorithms, or tools used in your project.
- Provide a high-level overview of your project's architecture or workflow.

3) Results:

- Present the output of your project.
- Share any data or experimental results you have obtained.
- Run product demo.

4) Conclusion:

- Summarize the key takeaways from your presentation.
- Discuss the significance of your outcomes in relation to the problem statement.
- Mention any insights or lessons learned during the project.

5) Future Directions:

- Suggest potential areas for further development or research.
- Share ideas for extending or improving your project.
- Identify any unanswered questions or limitations that could be addressed in the future.

ASSESSMENT RUBRIC

Criteria	Share	Quality level			
		very good	good	pass	failed
		10 – 8.5	8.4 – 7.0	6.9 – 5.0	4.9 – 0.0
Create object	20%	draw object with correct dimension	-	-	can not draw the object
Mathematical Control	20%	object move exact the same as requirement	-	-	cannot make object move or direction with mouse and keyboard
Visual Realism	20%	Correct antialiasing and global illumination shadow	-	-	cannot apply lighting and shadow
Integration & Logic	10%	The code is organized; variables are named clearly; modules talk to each other.	-	-	Code is a mess
Answer question	30%	Answer all questions	-	-	Answer < 1/3 of all questions